EMERGENCY ENROUTE · JURY NOTES

EVERY NUMBER ON SCREEN, AND WHERE IT COMES FROM

Plain-words notes for the prototype. Each value links to its source. A hollow circle ○ means a proxy (real data from a similar place) or an assumption (no public data found), exactly as on the site.

THE IDEA IN 30 SECONDS

- **One extra light.** Each traffic signal gets a **blue light** that switches on only when an ambulance is coming.
- **n+1.** The junction the ambulance is approaching *and the next one* react together. The first blinks blue and goes green for the ambulance. The next turns green early with a steady blue light, so its queue has drained by the time the ambulance gets there.
- **When it switches.** The ambulance sends its GPS position and speed every couple of seconds. The junction turns green when the ambulance is **94 s away** S14, or as a backup when it is **within 760 m** S14 ○, whichever comes first.
- **Why it matters.** A siren reaches only about **12 m** into a car at a junction S13. The blue light warns drivers hundreds of metres earlier, and the early green clears the queue the ambulance would otherwise be stuck behind.
- **No lane rule.** We dropped the middle-lane idea. All lanes keep moving; the gain comes from the early greens.

> **SAY IT LIKE THIS**
>
> Drivers hear a siren with seconds to react. Our blue light gives them a minute and a half, and turns the next two signals green so the queue is gone before the ambulance arrives.

HOW TO READ THE SCREEN

WHAT YOU SEE	WHAT IT MEANS
`4.8 km/h · S2`	A number and its **source tag.** S2 is a row in the sources list at the bottom of this page.

`760 m · S14 ○`	The **hollow circle** marks a proxy or an assumption.
Hover for 2 seconds	On the site, resting the pointer on any number or control for 2 s shows a small box explaining it in one sentence, with its source and the kind of number. It goes away when the pointer leaves.
`11:35 REAL TIME`	Real time since the ambulance appeared. **Playback runs 15× faster** than real life, so 11:35 of real time is about 46 s on screen. Speeds are never sped up, only the clock.
Without / With system	The same traffic, from the same random seed, replayed with the system off or on. Keyboard: `S`.
Journey bars	Journey time in each run on one scale. The J1 / J2 ticks show when the ambulance passed each junction.
Glass card, top right	What is happening right now, in two lines, and which step of the trip it is.
Red line with a number	Distance from the ambulance to the next junction, counted along the road. The same distance the trigger uses.
Blue light: blinking / steady / off	Blinking: **Alert**, green for the ambulance, red for everyone else. Steady: **Prewarn**, the next junction gone green early. Off: normal cycle.
Amber dots	General traffic, one dot per vehicle. Tight bright clusters are queues.

BANGALORE, HEAVY EVENING RAIN

Silk Board Junction, then the next signal towards Madiwala. The screen you asked about. Each row is one thing on that screen.

ON SCREEN	PLAIN MEANING	HOW IT IS WORKED OUT	KIND
4.8 km/h · S2	Average traffic speed in heavy rain.	Taken directly: Outer Ring Road after about 2 h of rain, 4.8 km/h with a 3.2 km queue S2. Cross-check: Silk Board's peak speed is 4.48 km/h even when dry S6.	SOURCED

24.0 km/h · S9	The fastest the ambulance aims to drive with the system.	Safe speed for 150 mm of water on the road, from the flood-speed formula S9: 24.26 km/h, stored as 24. The 150 mm depth is our assumption. It is a ceiling, not a speed reached: traffic ahead holds the ambulance to an average of **4.6 km/h.**	DERIVED
4.8 km/h · S2	The ambulance's speed with only a siren.	Our assumption: it moves no faster than traffic, because drivers ahead can't hear it in time S13 and most don't yield S17.	ASSUMPTION
760 m · S14	The signal also turns green once the ambulance is this close.	The distance a real Bengaluru pilot device used S14. A proxy because it is that device, not ours. In this run it fired at 759 m: crawling at about 5 km/h, the ambulance was about 9 minutes away, far over the 94 s rule, so the 760 m backup fired first.	PROXY
32 min 33 s	Journey time with only a siren.	Simulation result: 1,953 s to cover 1,712 m (1 km before Silk Board to 300 m past the next signal). About 10 minutes of it is spent fully stopped in Silk Board's queue.	SIMULATED
22 min 9 s	Journey time with the system.	Simulation result: 1,329 s over the same 1,712 m. Never stops. Close to the 21.4 min that pure 4.8 km/h would take.	SIMULATED
10 min 24 s	Time saved.	1,953.0 – 1,329.4 = 623.6 s. Marked “estimated” because it comes from the simulation, not a road test.	SIMULATED
J1 / J2 ticks	When the ambulance passed each junction.	Without: 23:55 and 28:52. With: 12:51 and 17:41.	SIMULATED
285 s cycle	Silk Board takes 285 s to give every road its	Measured timing: BTM 150 s, HSR 90 s, Hosur Road 45 s each side S5. The ambulance's road gets **45 s green, 240**	SOURCED

green once. **s red** every cycle.

Where the 10 minutes come from

WITHOUT THE SYSTEM

32 min 33 s

Stuck behind cars waiting out Silk Board's 240 s red. About **587 s fully stopped**. Average 3.2 km/h.

WITH THE SYSTEM

22 min 9 s

Silk Board turns green 760 m ahead, so the queue drains first. **0 s stopped**. Average 4.6 km/h.

BENCHMARK

21 min 24 s

1,712 m at a steady 4.8 km/h. With the system the ambulance gets within a minute of it.

> **SAY IT LIKE THIS**
>
> Almost all of the 10 minutes is the wait at Silk Board's red light. The ambulance's road gets 45 seconds of green in a 285-second cycle. With the system the signal goes green 760 m ahead, so the queue has cleared before the ambulance gets there.

The ambulance never obeys a red light in either run. The siren-only delay comes from the cars in front of it that do.

THE OTHER SCENARIOS

SCENARIO	TRAFFIC SPEED	WITHOUT → WITH	SAVED	WHAT IT SHOWS
Gurugram, Monday 6 pm Rajiv Chowk → Subhash Chowk	**18.9 km/h · S3** ○	**6:08 → 4:55**	**1 min 13 s**	Evening peak. New Delhi's evening-rush speed used as a proxy, since TomTom has no Gurugram index S3. Here the **94 s rule** fires first, at 894 m. because the

				ambulance is doing 34 km/h. Its 35 km/h aim is an assumption, conservative next to a real green corridor that did 12 km in 6 min S4.
Night, clear Hosur Road (control)	**29 km/h · S1**	**3:32 → 3:32**	**0 s**	The control case. With almost no traffic there is no queue to clear, so the system changes nothing. That is the honest result, and it shows the saving comes from queues, not from a trick. Speed derived from TomTom's Bengaluru figures S1; volume 25% of the Silk Board count (assumption).
Gurugram, monsoon waterlogging	**≈ 5 km/h · S9**	**34:09 → 31:50**	**2 min 19 s**	The limit case. At an assumed 250 mm of water the flood formula caps **everyone**, the ambulance included, at about 5 km/h S9. The system can't make the road passable. Real stories of 5-minute trips taking an hour S19.

EVERY CALCULATION, STEP BY STEP

Flood speed (the 24 km/h)

```
v(w) = 0.0009·w² − 0.5529·w + 86.9448      w = water depth (mm), v = safe speed (km/h)
v(150) = 0.0009 × 22,500 − 0.5529 × 150 + 86.9448
       = 20.25 − 82.94 + 86.94
       = 24.26 km/h   → stored as 24
At 300 mm or more the road counts as impassable (v = 0)
```

At 300 mm or more the road counts as impassable (v = 0).

From Pregnolato et al. (2017) S9. Table: 0 mm → 86.9, 100 mm → 40.7, 150 mm → 24.3, 250 mm → 5.0, 300 mm → 0 km/h.

The 94 s trigger

```
29 km/h ÷ 3.6 = 8.06 m/s           (clear-night Bengaluru speed, S1)
760 m ÷ 8.06 m/s = 94.3 s          (the pilot's 760 m, S14)
ETA = distance ÷ speed             (checked on every GPS packet)
Turn green when ETA ≤ 94.3 s  OR  distance ≤ 760 m
```

94.3 s is the warning the real pilot gives on a clear road. The 760 m backup covers an ambulance stuck in a jam: its speed is near zero, so its ETA never drops.

Unit conversions

```
km/h → m/s : divide by 3.6        4.8 km/h = 1.33 m/s     24 km/h = 6.67 m/s
seconds → min:s                    1,953 s = 32 min 33 s   1,329 s = 22 min 9 s
Screen time = real time ÷ 15       22 min 9 s real ≈ 89 s on screen
```

Journey distance and the benchmark

```
1,000 m before J1 + 405 m J1→J2 + 7 m (half of J2's box) + 300 m after = 1,712 m
1,712 m ÷ 1.33 m/s = 1,284 s ≈ 21 min 24 s   (steady 4.8 km/h, no stops)
```

The 405 m between the junctions is measured on the OpenStreetMap stand-in streets S20.

Time saved

```
Saved = journey without − journey with = 1,953.0 − 1,329.4 = 623.6 s = 10 min 24 s
```

Silk Board traffic per road

```
Total 17,816 veh/h, BTM 8,300 veh/h                    (S5, 9:30–10:30 am count)
Left for the others: 17,816 − 8,300 = 9,516
Split by green time 90 : 45 : 45  →  HSR 4,758, Hosur N 2,379, Hosur S 2,379
```

A morning count, used for the evening as a proxy. The greens add to 330 s in a 285 s cycle, so BTM and HSR must overlap; we overlap them by 60 s.

How many lanes

```
Junction capacity (IRC): 525 × road width (m) passenger-car units per hour of green   (S
Hosur S: 2,379 veh/h × 0.606 PCU/veh × 285 s ÷ 45 s = 9,137 PCU per hour of green
9,137 ÷ 525 = 17.4 m  →  5 lanes of 3.5 m
```

0.606 is the average size of a vehicle in car units, from the vehicle mix S7 and car-unit values S10. So “5 lanes” means the least road that can carry the measured traffic in the measured green time. Read it as effective lanes: Indian traffic is not lane-disciplined.

Signal changes

```
Amber 3 s + all-red 2 s = 5 s between greens      (picked from IRC ranges 3–6 s and 1–2
```

Every switch to green, including the system's, goes through this 5 s change, so no road ever jumps straight from green to red.

HOW THE SIMULATION WORKS, IN PLAIN WORDS

- **Real scale.** Every speed, distance and timing is real. Only the playback clock is sped up 15×.
- **Car-following.** Each vehicle speeds up, follows and brakes using the Intelligent Driver Model, a standard traffic model S16. Every vehicle gets its own desired speed around the scenario speed; two-wheelers ride a little faster and buses a little slower.
- **Groups of 10.** To keep it fast, the engine moves traffic in groups of 10 vehicles of one type (assumption). On screen each group is drawn as its 10 vehicles.
- **Traffic in.** Vehicles arrive on each road at the hourly counts (S5 for Silk Board), divided by 10.
- **Junction capacity.** A green light lets traffic through no faster than the IRC capacity formula allows S11, so queues drain at a realistic rate.
- **Warm-up.** Before the ambulance appears the simulation runs three full signal cycles, so the queues are already there.

- **Fair comparison.** Both runs use the same random seed: identical traffic until the ambulance arrives.
- **The ambulance.** It ignores red lights in both runs and never overtakes. It is slowed only by the vehicles ahead in its lane.
- **Deterministic.** The same scenario always gives the same result, so scrubbing backwards shows exactly what happened.

HOW IT WOULD WORK IN THE REAL WORLD

Nothing has been field-tested. The prototype simulates what this hardware would do.

STEP	HARDWARE	IN THE PROTOTYPE
1. The ambulance reports where it is	GPS (NEO-M8N) and compass (MPU9250) send `{lat, lon, heading, speed, timestamp}` over 4G or LoRa.	A GPS packet every 1.5 s (assumption; the concept says every 1–2 s).
2. Junctions get the message	Through a central server, or junction to junction over MQTT.	Both junctions read every packet.
3. Each junction does the maths	Distance by the haversine formula; a heading check that the ambulance is coming towards it; ETA = distance ÷ speed.	Distance along the road; the ambulance is always on the route, so the heading check always passes.
4. The trigger fires	A Raspberry Pi or ESP32 switches a relay: blue light on, signal set to green for the ambulance.	ETA ≤ 94.3 s or distance ≤ 760 m.
5. The next junction is warned (n+1)	Junction N tells junction N+1, which turns its blue light on and starts clearing.	J2 goes green with a steady blue light at the same moment.
6. Reset	When the distance starts growing, or an ANPR camera confirms the ambulance passed.	When the back of the ambulance clears the far side of the junction; then amber, all-red, and the normal cycle carries on.
Backup	A microphone with a siren classifier, if GPS or the network drops.	Not simulated.

LIKELY QUESTIONS

- **Is 10 minutes realistic?**
- **Why does the ambulance only average 4.6 km/h when its limit is 24?**
- **Why does the night scenario save nothing?**
- **Where does 760 m come from?**
- **Why trigger on time and not just distance?**
- **Doesn't holding everyone else at red cause new jams?**
- **Why only 15× speed-up?**
- **Why is the Bangalore map not Silk Board?**
- **Why are cars in groups of 10?**

WHAT TO ADMIT BEFORE YOU'RE ASKED

1. **Stand-in maps.** The 3D streets in both cities are an OpenStreetMap stand-in, not the real junctions. The J1→J2 distances are measured on those streets.
2. **Unverified second junction.** “Next signal towards Madiwala” is unconfirmed, and its 150 s timing is an assumption.
3. **One rain observation.** 4.8 km/h is from one evening on the Outer Ring Road S2; Silk Board's own 4.48 km/h S6 supports it.
4. **Morning count for an evening scene.** Silk Board's volumes are a 9:30–10:30 am count S5.
5. **Assumed water depth.** 150 mm (Bangalore) and 250 mm (Gurugram) are assumptions.
6. **No overtaking.** Our ambulance never passes traffic, so the saving is conservative.
7. **Estimates.** All journey times and savings come from the simulation, not a field trial.

SOURCES

ID	WHAT IT GIVES US	SOURCE
S1	Bengaluru 2025: 74.4% congestion, 36:09 per 10 km, evening peak 13.2 km/h	TomTom Traffic Index – Bengaluru
S2	Outer Ring Road after about 2 h of rain: 4.8 km/h, 3.2 km queue	Deccan Herald – Bumper-to-bumper traffic on ORR after evening showers
S3	New Delhi 2025: evening rush 18.9 km/h, 192% congestion at 6 pm	TomTom Traffic Index – New Delhi
S4	IGI → Medanta green corridor: 12 km in 6 min	The Tribune – 12-km green corridor to Medanta
S5	Silk Board: 17,816 veh/h, 8,300 from BTM, 285 s cycle, greens 150 / 90 / 45 / 45 s	Deccan Herald – How Silk Board sinks under vehicle movement
S6	Silk Board average peak speed 4.48 km/h (2017 study)	Wikipedia – Silk Board junction
S7	Bengaluru fleet: 1.28 crore vehicles, 87 lakh two-wheelers, 25.2 lakh cars	Deccan Herald – Bengaluru has the most private cars
S8	Registered vehicles by type, Bengaluru 2024-25	OpenCity – Total Vehicles Registered in Bengaluru
S9	Flood depth vs safe speed: $v(w) = 0.0009w^2 - 0.5529w + 86.9448$; impassable at 300 mm	Pregnolato et al. (2017), *Transportation Research Part D* 55. doi:10.1016/j.trd.2017.06.020
S10	Car-unit values: two-wheeler 0.46, auto 0.78, bus 2.2	Sali & Ajitha – Delay and saturation flow (ICCESP 2024)
S11	IRC junction capacity: 525 × road width (m) car units per hour of green	NPTEL – Traffic signal design
S12	IRC:93-1985: cycle 40–120 s, amber 3–6 s, all-red 1–2 s	InfraLens – IRC 93-1985 summary

S13	A siren reaches about 12 m into a car at a junction; reaction time about 1.6 s	Fire Engineering – Siren limitations and intersection crashes
S14	Bengaluru pilot: the signal turns green when the ambulance is 760 m away	Deccan Herald – A device to give ambulances right of way
S15	Bengaluru: about 3,000 ambulance trips a day; ₹10,000 fine under MV Act 194E	Deccan Herald – No green for the ambulance
S16	Intelligent Driver Model (car-following)	Treiber, Hennecke & Helbing (2000), *Phys. Rev. E* 62. arXiv:cond-mat/0002177
S17	Indian cities: 25–30 min response; 62% of ambulance drivers say motorists don't yield	The Hindu via Vision IAS – Reclaiming the Golden Hour
S18	Gurugram underpass speed limit 30 km/h	The Tribune – 30 km/h limit for 9 underpasses
S19	Gurugram rain: a 5–7 min trip took 1.5	Business Today – Gurugram rains leave